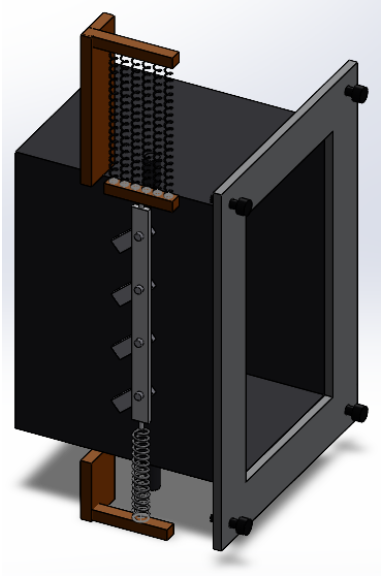
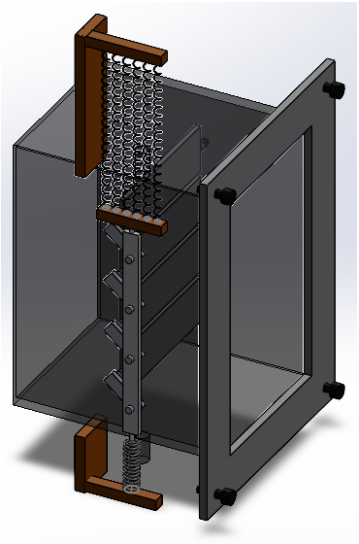
SMA Brake-Duct Actuator Module

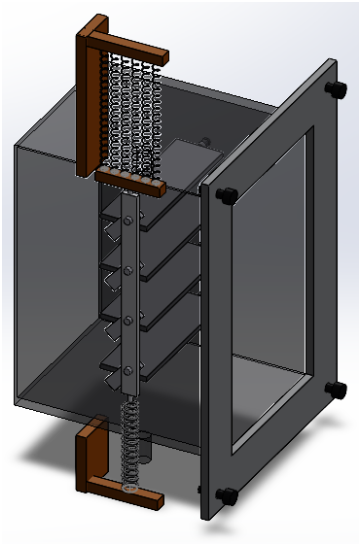
Passive, temperature-driven louver actuator for brake cooling: design, simulation and test-rig package. Prototype design, fabrication-ready; not yet built.



Module (SOLIDWORKS)



Cold: louvers closed



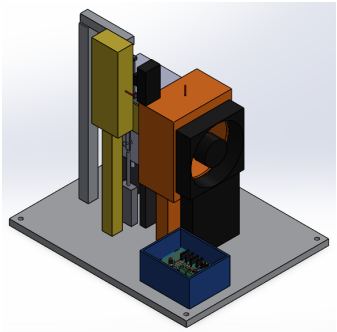
Hot: louvers open 88.5°

Specification

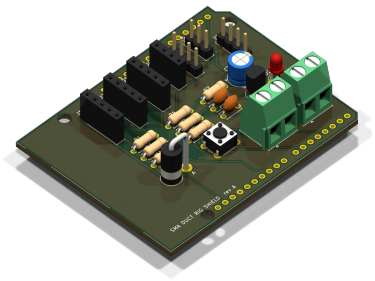
Function	Opens brake-cooling louvers when the brakes are hot, closes them on straights. No electronics, no power.
Actuator	6 x NiTiCu springs, 0.5 mm wire, 2.5 mm coil, 19 turns, black high-emissivity coating, steel bias spring
Trigger	Opens 72-75 °C, closes 63-60 °C (hysteresis 11.5 °C)
Output	15.15 mm rod stroke, louvers 0 to 88.5°, 33.9 N pack force (MAPDL)
Response	Opens in 3.6 s, closes in 1.9 s (1.2 mm wire: 5.7 s to close)
Structure	All stress checks pass, lowest factor of safety 3.3 (ANSYS MAPDL)
Envelope	89 x 182 x 94 mm module: CFRP throat, 4 titanium louvers, crank linkage, Al inlet flange (4 x M3)

Validation

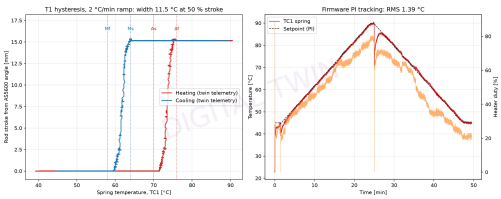
Thermal CFD	Rotor radiation at the spring pocket 119-129 kW/m² (Fluent DO); air alone cannot trigger the spring
Lap model	Opens 4-5 s after the 3 hardest braking zones, open 2 % of straight-line time
Aero CFD	Isolated wheel at 300 km/h, 3 meshes: any drag saving is capped at 1.1-1.4 % by the duct's momentum flux (brief claimed 18.4 %); the direct difference is within solver noise
Bench rig	Lamp + blower rig, Arduino shield (ERC/DRC clean, Gerbers), firmware 57 % flash / 55 % RAM
Digital twin	Same firmware code + physics: reproduces the car lap's openings (87 % agreement); 1,000 cycles, 0 faults; caught 3 design errors before build



Bench rig (SOLIDWORKS)



Sensor shield (KiCad 3D)



Digital-twin hysteresis test

Tools: SOLIDWORKS 2025, ANSYS Fluent and MAPDL (Student 2026 R1), KiCad 10, Arduino (arduino-cli), C++/MSVC, Python. Drawing SMA-DM-001, CAD, Gerbers, firmware and test scripts in the project repository. Rig and twin results are predictions pending a physical build.